Ricardian Model of Comparative Advantage with Bayesian Uncertainty Quantification

A project submitted
for partial fulfillment of the requirements for the award of the degree of
Master of Science

by

MAMIDALA YASHWANTH

M.Sc. in Statistics & Computing
Exam Roll Number : 24419STC030

Under the supervision of

Dr. Akanksha Gupta
Department of Statistics, BHU

Centre for Interdisciplinary Mathematical Sciences
I.Sc., Banaras Hindu University
Varanasi, India-221005

Enrollment No: 482736 May, 2026

Declaration

I, **Mamidala Yashwanth**, hereby declare that the project titled ***Ricardian Model of Comparative Advantage with Bayesian Uncertainty Quantification*** and the work presented therein are my own and original, carried out under the supervision of **Dr. Akanksha Gupta, Department of Statistics**, Banaras Hindu University, during the period from January 2026 to May 2026. This project has been submitted to the DST-Centre for Interdisciplinary Mathematical Sciences, Banaras Hindu University, in partial fulfilment of the requirements for the M.Sc. in Statistics & Computing course. To the best of my knowledge and belief, the information and data presented in this project are authentic. This work has not been submitted to any other university or institution for the award of any degree, diploma, or fellowship, nor has it been published previously.

Place: Varanasi **Mamidala Yashwanth**

Date: 08/05/2026 **24419STC030**

CERTIFICATE

This is to certify that **Mamidala Yashwanth**, a student of the **Master of Science in Statistics & Computing**, Semester IV of the **DST-Centre for Interdisciplinary Mathematical Sciences**, Institute of Science, Banaras Hindu University, bearing exam roll no. **24419STC030** and enrollment no. **482736** of 2024-2026 has successfully completed his project entitled "**Ricardian Model of Comparative Advantage with Bayesian Uncertainty Quantification**". The project was completed under my direct supervision and guidance for the partial fulfillment of the course curriculum of Master of Science in Statistics & Computing. He carried out the whole work with utmost sincerity, and the present report is the outcome of his sincere efforts and endeavors. This work either in the present form or in part in any other form is not being submitted anywhere to the best of my knowledge.

Supervisor

Course Coordinator

Coordinator, DST-CIMS

Acknowledgment

I am honoured to have this opportunity to convey my sincere gratitude to my supervisor, **Dr. Akanksha Gupta**, for her invaluable guidance, constant motivation, and caring mentorship throughout the course of this research. Without her immense knowledge and insightful vision, this work would not have taken its present shape.

I would also like to express my sincere thanks to the Ph.D. scholar, **Mr.Pranjal Kumar Pandey**, for their continuous support and valuable suggestions, which greatly helped me in better understanding and improving various aspects of this project.

Place: Varanasi **Mamidala Yashwanth**

Date:08/05/2026 **24419STC030**

Abstract

The Ricardian Model of Comparative Advantage explains why nations specialise and trade based on relative productivity differences rather than absolute superiority. However, its classical formulation treats agricultural yields as fixed point estimates, ignoring year-to-year variability driven by weather, technology, and policy shocks. This project applies the Ricardian Model to two bilateral trade scenarios using 2024 FAOSTAT data – China vs India (Rice and Wheat) and USA vs India (Maize and Rice) – computing productivity, opportunity costs, Production Possibility Frontiers, and Terms of Trade. Model 1 reveals no meaningful comparative advantage between China and India due to near-identical opportunity costs. Model 2 identifies clear comparative advantage – USA in Maize and India in Rice – with Terms of Trade between 0.771 and 1.217. The project then extends Model 2 through a Bayesian uncertainty quantification framework. Using 53 years of historical yield data (1961–2013) as prior and 11 recent observations (2014–2024) as likelihood, Log-Normal priors are constructed on detrended log-scale residuals and updated via Normal-Normal conjugate inference. Uncertainty is propagated through Monte Carlo simulation with 100,000 draws, yielding a posterior probability of 100% that India holds comparative advantage in Rice. The results confirm that the trade conclusion is statistically robust under all realistic agricultural yield uncertainty, bridging classical trade theory with modern Bayesian statistical inference.

Keywords: Ricardian Model, Comparative Advantage, Opportunity Cost, Production Possibility Frontier, Terms of Trade, Bayesian Inference, Log-Normal Prior, Normal-Normal Conjugate, Monte Carlo Simulation, FAOSTAT, Agricultural Yield, Uncertainty Quantification, International Trade Theory.

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1 Introduction

International trade has been a cornerstone of economic development since the early nineteenth century. Nations engage in trade not because they are forced to, but because specialisation and exchange allow each country to consume beyond its own production capabilities. The theoretical foundation for this phenomenon was laid by **David Ricardo** in **1817**, whose principle of Comparative Advantage remains one of the most elegant and enduring ideas in all of economics. Unlike Adam Smith's earlier notion of absolute advantage – which suggested that trade only benefits countries that are more productive in a given good – Ricardo demonstrated that trade is beneficial even when one country is more productive in every good, provided that opportunity costs differ between nations.

1.1 Background

The Ricardian Model of Comparative Advantage provides a framework for understanding why countries specialise in the production of certain goods and import others. The model is built on the concept of opportunity cost – the amount of one good that must be foregone in order to produce an additional unit of another. A country holds comparative advantage in a good when its opportunity cost of producing that good is lower than that of its trading partner. Under this framework, both countries gain from trade by specialising in their respective comparative advantage goods and exchanging at an agreed Terms of Trade lying between their domestic opportunity cost ratios.

In the context of agricultural commodities, comparative advantage has significant implications for food security, rural employment, and trade policy. Countries such as India and the United States of America are among the world's largest producers of staple crops including Rice and Maize, and their bilateral trade patterns have been shaped by underlying productivity differences over decades. Understanding whether these productivity differences constitute a genuine and statistically robust basis for trade is therefore both theoretically and policy-relevant.

1.2 Motivation

While the classical Ricardian framework is powerful, it operates on a fundamental assumption that agricultural yields are fixed and deterministic. In practice, crop yields fluctuate significantly from year to year due to monsoon variability, pest and disease outbreaks, input availability, irrigation infrastructure, and government policy interventions. Treating a single year's yield observation as the definitive measure of a country's productive capacity ignores this inherent uncertainty and may lead to trade conclusions that are sensitive to anomalous observations.

This project is motivated by the question: *Even after accounting for realistic agricultural yield uncertainty, how confident can we be that the Ricardian trade conclusion holds?* To answer this, the classical model is extended using a Bayesian statistical framework that replaces fixed point estimates with probability distributions and produces a formal, probability-qualified statement of comparative advantage.

1.3 Objectives

The objectives of this project are as follows:

1. To apply the classical Ricardian Model of Comparative Advantage to two bilateral trade scenarios – China vs India (Rice and Wheat) and USA vs India (Maize and Rice) – using 2024 FAOSTAT production data.

2. To compute productivity, opportunity costs, Production Possibility Frontiers, and Terms of Trade for each model and determine whether a theoretical basis for mutually beneficial trade exists.

3. To extend Model 2 (USA vs India) through a Bayesian uncertainty quantification framework using 64 years of historical yield data (1961–2024), formally quantifying confidence in the comparative advantage conclusion.

4. To construct Log-Normal priors from detrended historical yield residuals (1961–2013), update them with recent observations (2014–2024) via Normal-Normal conjugate inference, and propagate uncertainty through Monte Carlo simulation to obtain posterior comparative advantage probabilities.

1.4 Data Source

All data used in this project are sourced from the Food and Agriculture Organization of the United Nations (FAO) statistical database, FAOSTAT, available at `https://www.fao.org/faostat/en/#data`. FAOSTAT provides comprehensive country-level data on crop production (tonnes) and area harvested (hectares) for all major agricultural commodities across all nations from 1961 to the present. Yield (productivity in tonnes per hectare) is derived as the ratio of production to area harvested. Data for the classical models uses the 2024 observation year, while the Bayesian extension uses the full historical series from 1961 to 2024 for India and the United States of America across Maize and Rice.

2 The Ricardian Model — Theory and Background

2.1 Historical Context

David Ricardo, in his 1817 work *Principles of Political Economy and Taxation*, introduced the Law of Comparative Advantage. He demonstrated — through the famous England–Portugal example involving cloth and wine — that even if one country is absolutely more efficient in producing all goods, both countries can gain from trade by specialising in goods where they have a lower relative opportunity cost.

2.2 Core Theoretical Framework

The Ricardian Model is built on the following core ideas:

- Labour (or land, in an agricultural context) is the only factor of production.
- Productivity differences between countries are the sole source of comparative advantage.
- Within a country, resources can move freely between sectors.
- Trade leads to specialisation: each country produces more of the good in which it has a comparative advantage.
- Both countries gain from trade even if one is absolutely more productive in all goods.

2.3 Assumptions of the Ricardian Model

The classical Ricardian Model rests on the following assumptions:

- There are only two countries and two goods.
- There is only one factor of production (labour or land).
- Technology and productivity are fixed and constant across production levels (constant returns to scale).
- Factors of production are perfectly mobile within a country but immobile between countries.
- Markets are perfectly competitive — no monopolies and no market distortions.
- There are no transportation costs or trade barriers.
- Full employment of resources is assumed.

- The Production Possibility Frontier (PPF) is linear (a straight line) because opportunity cost is constant — a direct consequence of the single-factor and constant productivity assumptions.

2.4 Key Concepts

2.4.1 Absolute Advantage

A country has an absolute advantage in a good if it can produce more of that good per unit of resource than another country. In the Ricardian Model, absolute advantage alone does not determine trade. Even if Country A has an absolute advantage in all goods, trade can still be beneficial.

2.4.2 Opportunity Cost

The opportunity cost of producing one unit of Good X is the amount of Good Y that must be sacrificed. It is calculated as:

$$\text{OC of } X \text{ (in terms of } Y) = \frac{\text{Productivity of } Y}{\text{Productivity of } X}$$

A lower opportunity cost indicates that a country gives up less of the other good to produce a unit of X, implying a comparative advantage.

2.4.3 Comparative Advantage

A country has a comparative advantage in Good X if its opportunity cost of producing X is lower than that of another country. The country should specialise in and export the good in which it has a comparative advantage.

2.5 Production Possibility Frontier (PPF)

The Production Possibility Frontier (PPF) represents all possible combinations of two goods that a country can produce given its total resources and technology.

Under the Ricardian Model:

- The PPF is a straight line because opportunity cost is constant.
- The slope of the PPF equals the negative of the opportunity cost ratio.
- The X-intercept represents the maximum possible production of Good X (when all resources are devoted to X).

- The Y-intercept represents the maximum possible production of Good Y (when all resources are devoted to Y).
- A country with a steeper PPF has a higher opportunity cost of the good on the X-axis.
- Differences in PPF slopes between countries provide the basis for trade.

In this project, PPF endpoints are derived by normalising total land to 100 units and multiplying by the respective productivity values. The slope of each PPF directly corresponds to the opportunity cost calculated in the analysis.

2.6 Bayesian Inference in Economics

Bayesian statistical inference, rooted in Bayes' theorem (1763), provides a principled framework for updating prior beliefs in light of new evidence. Unlike classical frequentist methods, which treat parameters as fixed unknown constants, the Bayesian approach treats parameters as random variables with probability distributions that reflect uncertainty. The posterior distribution – combining prior belief with observed likelihood – serves as the complete inferential summary of parameter uncertainty.

The application of Bayesian methods in economics has grown substantially over the past three decades. Geweke (1999) provided an influential review of Bayesian simulation methods in econometrics, establishing the foundations for Bayesian analysis of economic models. Sims and Zha (1998) demonstrated the advantages of Bayesian Vector Autoregression for macroeconomic forecasting, showing that incorporating prior information improves out-of-sample prediction.

In the context of conjugate inference, the Normal-Normal conjugate pair – where a Normal prior on the mean parameter combined with a Normal likelihood yields a Normal posterior – is one of the most widely used frameworks in applied Bayesian statistics. DeGroot (1970) provides a comprehensive treatment of conjugate families and their properties. The conjugate approach is particularly attractive in applied settings because it yields closed-form posterior distributions, avoiding the need for computationally intensive Markov Chain Monte Carlo methods.

Bayesian methods have been increasingly applied to questions of trade and economic policy. Koop and Korobilis (2010) demonstrated the advantages of Bayesian model averaging in trade policy analysis, while Zellner (1996) established foundational results for Bayesian inference in structural economic models. The use of informative priors derived from historical data – as employed in this project – is consistent with the empirical Bayes tradition pioneered by Robbins (1956).

2.7 Agricultural Yield Modelling

Agricultural yield modelling has a rich history in both agronomy and agricultural economics. Yields – measured as output per unit of cultivated area – are determined by a complex interaction of climatic, biological, technological, and policy factors. Statistical modelling of yield distributions is essential for risk assessment, food security planning, and trade policy analysis.

The Log-Normal distribution has been widely adopted as a standard model for agricultural yield data. Gallagher (1987) demonstrated that crop yield distributions across farm-level observations are well approximated by the Log-Normal distribution, owing to the multiplicative nature of yield-determining factors. Day (1965) provided early evidence that county-level yield distributions in the United States follow approximately Log-Normal patterns. More recently, Tack, Harri, and Coble (2012) confirmed the Log-Normal assumption for Maize yields using modern farm-level data.

The presence of long-run upward trends in agricultural yields – driven primarily by the Green Revolution of the 1960s and 1970s and subsequent technological improvements – has been extensively documented. Evenson and Gollin (2003) estimated that Green Revolution technologies increased crop yields by approximately 1 percent per year across major cereal crops in developing countries over the second half of the twentieth century. Fuglie (2012) documented continued total factor productivity growth in agriculture through the 2000s, confirming the persistence of technological trend in yield data.

The importance of detrending yield series before distributional modelling has been recognised in the agricultural risk literature. Sherrick et al. (2004) demonstrated that failure to account for trend in yield data leads to overestimation of yield variance and distorted risk assessments. This motivates the detrending procedure adopted in the present project, where Ordinary Least Squares regression on the log scale is used to isolate genuine year-to-year variability from long-run productivity growth.

Monte Carlo simulation methods for uncertainty propagation in agricultural models have been widely applied. Richardson, Schumann, and Feldman (2008) demonstrated the use of Monte Carlo simulation in farm-level risk analysis, showing that simulated distributions of economic outcomes accurately reflect underlying yield uncertainty. The application of Monte Carlo methods to comparative advantage analysis – as employed in this project – represents a natural extension of this tradition to the domain of international trade theory.

3 Data Description

All data used in this project are sourced from the Food and Agriculture Organization of the United Nations statistical database, FAOSTAT, available at https://www.fao.org/faostat/en/#data. FAOSTAT provides country-level annual data on crop production (tonnes) and area harvested (hectares) for all major agricultural commodities worldwide.

3.1 Model 1 – China vs India (Rice & Wheat)

The first dataset covers China and India for Rice and Wheat for the observation year 2024. Only four variables are retained for analysis:

- **Area** – Country name (China, India)
- **Element** – Area Harvested (hectares) or Production (tonnes)
- **Item** – Crop name (Rice, Wheat)
- **Value** – Numerical value of the element

Table 1 presents the raw data values used in Model 1.

Table 1: Raw Data – Model 1: China vs India (2024)

Country	Crop	Area Harvested (ha)	Production (tonnes)
China	Rice	28,700,000	203,610,000
China	Wheat	23,613,000	139,330,000
India	Rice	50,511,930	217,867,900
India	Wheat	31,400,000	113,290,000

3.2 Model 2 – USA vs India (Maize & Rice)

The second dataset covers the United States of America and India for Maize and Rice for the observation year 2024. The same four variables are retained. Table 2 presents the raw data values used in Model 2.

Table 2: Raw Data – Model 2: USA vs India (2024)

Country	Crop	Area Harvested (ha)	Production (tonnes)
USA	Maize	33,593,000	389,690,000
USA	Rice	1,016,000	9,095,000
India	Maize	10,130,000	35,960,000
India	Rice	50,511,930	217,867,900

3.3 Historical Data – Bayesian Extension

The Bayesian extension uses a longitudinal dataset covering India and the United States of America for Maize and Rice spanning 1961 to 2024 – a total of 64 years of annual observations. The dataset contains 504 records with the same four variables as above, additionally including a Year column.

The dataset is divided into two windows based on agricultural era:

- **Prior window (1961–2013):** 53 years of historical data representing the long-run productivity trend from the pre-Green Revolution era through early modern farming
- **Likelihood window (2014–2024):** 11 years of recent data representing the modern precision farming era used as observed evidence to update prior beliefs

Table 3 presents descriptive statistics of annual yields for both windows across all four country-crop pairs.

Table 3: Summary Statistics of Annual Yields (t/ha) – Prior vs Likelihood Window

Country	Crop	Window	Mean	Std	Min	Max	Skewness
India	Maize	Prior (1961–2013)	1.503	0.497	0.724	2.876	0.797
India	Maize	Likelihood (2014–2024)	3.037	0.315	2.589	3.550	0.329
India	Rice	Prior (1961–2013)	2.390	0.697	1.123	4.059	0.245
India	Rice	Likelihood (2014–2024)	3.977	0.166	3.590	4.220	-0.608
USA	Maize	Prior (1961–2013)	7.051	1.872	3.928	11.232	0.074
USA	Maize	Likelihood (2014–2024)	10.976	0.308	10.613	11.464	-0.743
USA	Rice	Prior (1961–2013)	6.138	1.237	3.800	8.801	0.091
USA	Rice	Likelihood (2014–2024)	8.460	0.160	8.143	8.953	-0.396

The likelihood window consistently shows higher mean yields and lower variability than the prior window across all four pairs, confirming continued productivity growth and greater stability in modern agricultural production. All graphs, distributional analysis, and trend visualisations derived from this dataset are presented in the Methodology and Results sections.

4 Methodology

This project employs a two-stage analytical framework. The first stage applies the classical Ricardian Model of Comparative Advantage to two bilateral trade scenarios using 2024 FAOSTAT data. The second stage extends the analysis through a Bayesian uncertainty quantification framework applied to Model 2 (USA vs India), incorporating 64 years of historical yield data to produce probability-qualified trade conclusions.

4.1 Classical Ricardian Model

Productivity

For each country-crop pair, productivity (yield) is computed as output per unit of cultivated land:

$$\text{Productivity}_i^j = \frac{\text{Production}_i^j \text{ (tonnes)}}{\text{Area Harvested}_i^j \text{ (hectares)}}$$

where i denotes the country and j denotes the crop.

Opportunity Cost

The opportunity cost of producing one unit of good j is the amount of good k that must be foregone. For country i:

$$OC_i^j = \frac{\text{Productivity}_i^k}{\text{Productivity}_i^j}$$

The country with the lower opportunity cost in good j holds comparative advantage in that good and should specialise in it.

Production Possibility Frontier

The Production Possibility Frontier (PPF) represents all combinations of two goods a country can produce by fully allocating its productive resources. For a single unit of land normalised to 1:

$$\text{PPF}_i : \quad \frac{Q_i^j}{\text{Productivity}_i^j} + \frac{Q_i^k}{\text{Productivity}_i^k} = 1$$

The slope of the PPF equals the negative opportunity cost ratio. Countries with different PPF slopes have different opportunity costs and thus a basis for trade.

Terms of Trade

The Terms of Trade (ToT) define the international exchange rate at which trade is mutually beneficial. For trade to benefit both countries, the ToT must lie strictly between the two domestic opportunity cost ratios:

$$OC^{j}_{\text{country A}} < \text{ToT} < OC^{j}_{\text{country B}}$$

The midpoint of this range is used as the suggested Terms of Trade in this project.

Absolute vs Comparative Advantage

It is important to distinguish between absolute and comparative advantage. A country holds absolute advantage in a good if its productivity is higher than its trading partner. A country holds comparative advantage in a good if its opportunity cost is lower than its trading partner. Trade is driven entirely by comparative advantage – not absolute advantage. A country may hold absolute advantage in every good and still benefit from trade by specialising in its comparative advantage good.

4.2 Bayesian Extension

The classical Ricardian model treats yields as fixed point estimates, ignoring year-to-year variability. The Bayesian extension addresses this by modelling yields as random variables with probability distributions, and formally quantifying the confidence in the comparative advantage conclusion. The framework follows Bayes' theorem:

$$P(\theta \mid x) = \frac{P(x \mid \theta) \cdot P(\theta)}{P(x)}$$

where θ represents the true yield, $P(\theta)$ is the prior distribution, $P(x \mid \theta)$ is the likelihood, $P(\theta \mid x)$ is the posterior distribution, and $P(x)$ is the normalising constant (evidence) handled implicitly by the conjugate update.

4.2.1 Prior Construction

Distributional Choice. Exploratory analysis of prior window yields (1961–2013) using histograms and Kernel Density Estimation plots revealed right-skewed distributions for all four country-crop pairs. This is consistent with the multiplicative nature of agricultural yield growth – yields improve by percentage increments each year rather than fixed absolute amounts. A Log-Normal distribution is therefore adopted as the prior:

$$Y \sim \text{LogNormal}(\mu_{log}, \sigma^2_{log})$$

Taking the natural logarithm transforms yields to the Normal scale:

$$\log(Y) \sim \mathcal{N}(\mu_{log}, \sigma^2_{log})$$

Figure 1 presents the KDE plots confirming that Log-Normal fits the prior window data better than Normal for all four pairs.

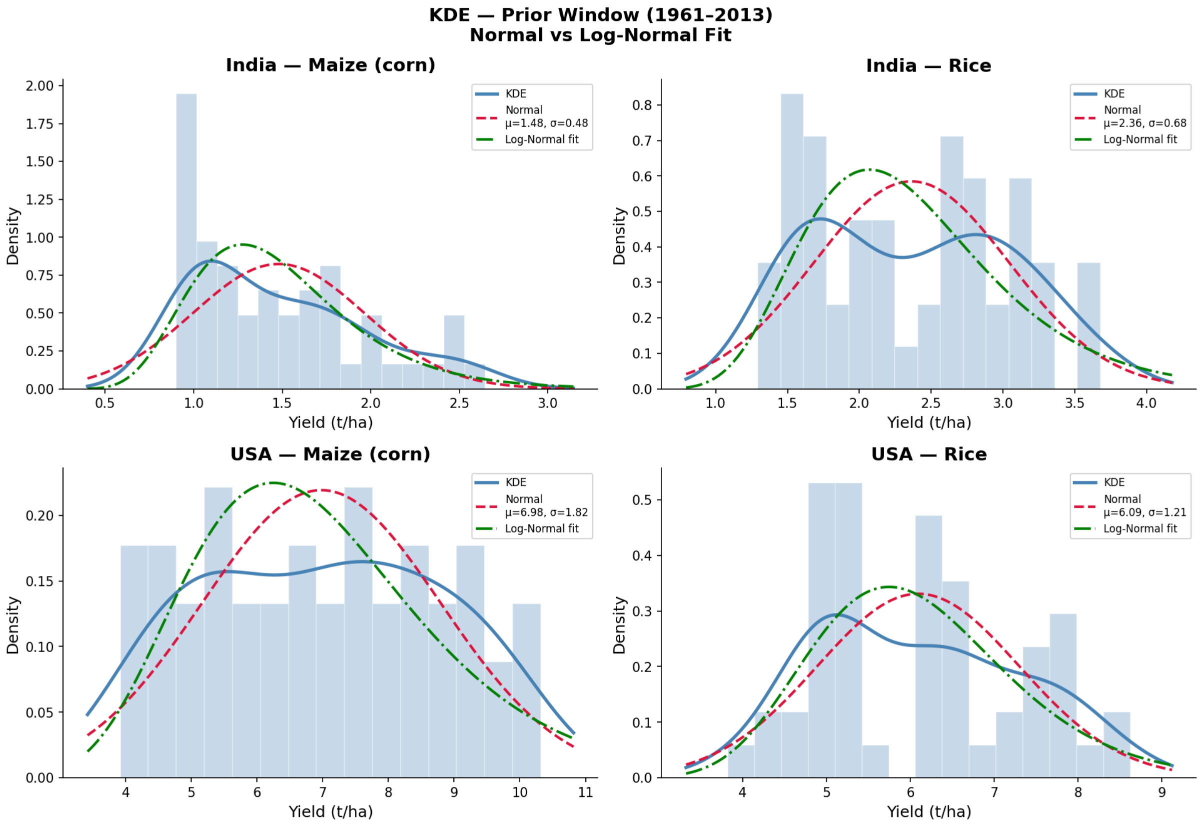

Figure 1: KDE Plots – Prior Window (1961–2013): Normal vs Log-Normal Fit

Detrending on Log Scale. Even after log transformation, a long-run upward trend remains in yields due to the Green Revolution and successive technological improvements. This trend artificially inflates variance and violates the stationarity assumption required for prior construction. The trend is removed using Ordinary Least Squares regression on the log scale:

$$\log(Y_t) = \alpha + \beta \cdot t + \epsilon_t$$

The residuals ϵ_t represent genuine year-to-year agricultural uncertainty – free from both skewness and trend – and form the basis for the prior distribution. Shapiro-Wilk normality tests confirm approximate normality of log-scale residuals for all four pairs, as shown in Figure 2.

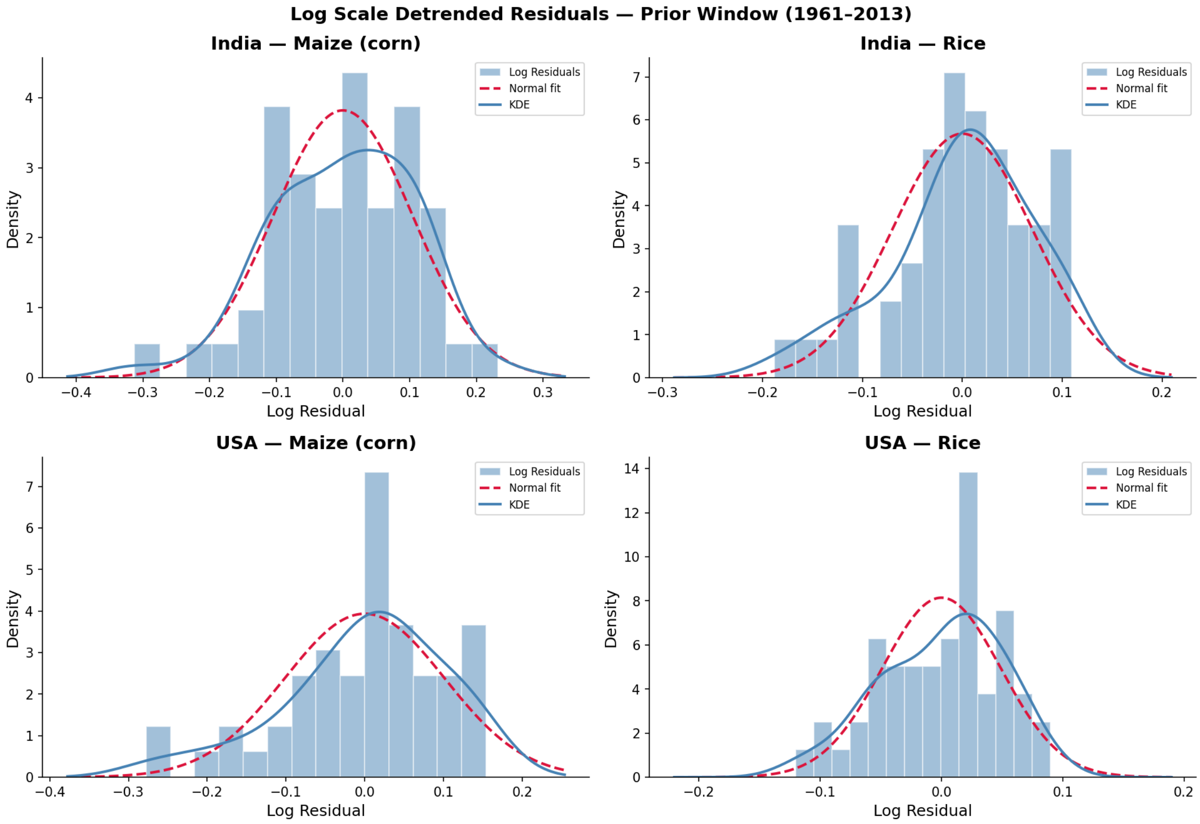

Figure 2: Log Scale Detrended Residuals – Prior Window (1961–2013)

Prior Parameters. The prior parameters on the log scale are set as follows:

- μ_{log}^{prior} = OLS trend-projected log yield at end of prior window (2013)
- σ_{log}^{prior} = standard deviation of log-scale detrended residuals (1961–2013)

Table 4 summarises the prior parameters for each country-crop pair.

Table 4: Prior Parameters on Log Scale – Prior Window (1961–2013)

Country	**Crop**	μ_{log}^{prior}	σ_{log}^{prior}	**Shapiro W**	**p-value**
India	Maize	0.8430	0.1045	0.9767	0.3832
India	Rice	1.3095	0.0703	0.9575	0.0568
USA	Maize (corn)	2.3389	0.1013	0.9491	0.0247
USA	Rice	2.1143	0.0490	0.9725	0.2583

4.2.2 Likelihood Construction

The likelihood window covers 2014 to 2024 – 11 annual yield observations per country-crop pair. Log yields in this window serve as the observed evidence. The likelihood for multiple observations is:

$$L(\mu) = \prod_{t=2014}^{2024} \mathcal{N}(\log Y_t \mid \mu, \sigma^2_{obs})$$

which simplifies to a Normal likelihood with sufficient statistic $\bar{x}_{log}$ – the sample mean of log yields in the likelihood window. The likelihood parameters are summarised in Table 5.

Table 5: Likelihood Parameters on Log Scale – Likelihood Window (2014–2024)

Country	**Crop**	**n**	$\bar{x}_{log}$	σ_{obs}
India	Maize	11	1.1115	0.1023
India	Rice	11	1.3822	0.0650
USA	Maize (corn)	11	2.3961	0.0300
USA	Rice	11	2.1356	0.0196

4.2.3 Posterior Update

The Normal-Normal conjugate update is applied on the log scale. Treating σ^2 as known and fixed at σ^2_{obs}, the closed-form posterior update formulas are:

$$\mu_{post} = \frac{\dfrac{\mu_0}{\sigma_0^2} + \dfrac{n \cdot \bar{x}_{log}}{\sigma^2_{obs}}}{\dfrac{1}{\sigma_0^2} + \dfrac{n}{\sigma^2_{obs}}}$$

$$\sigma^2_{post} = \frac{1}{\dfrac{1}{\sigma_0^2} + \dfrac{n}{\sigma^2_{obs}}}$$

where μ_0 and σ_0 are prior parameters, $\bar{x}_{log}$ and σ_{obs} are likelihood parameters, and $n = 11$ is the number of likelihood observations. The posterior mean is transformed back to the original scale as:

$$\mu_{original} = e^{\mu_{post} + \frac{\sigma^2_{post}}{2}}$$

Figure 3 presents the prior, likelihood, and posterior distributions on the log scale for all four country-crop pairs, and Table 6 summarises the posterior parameters.

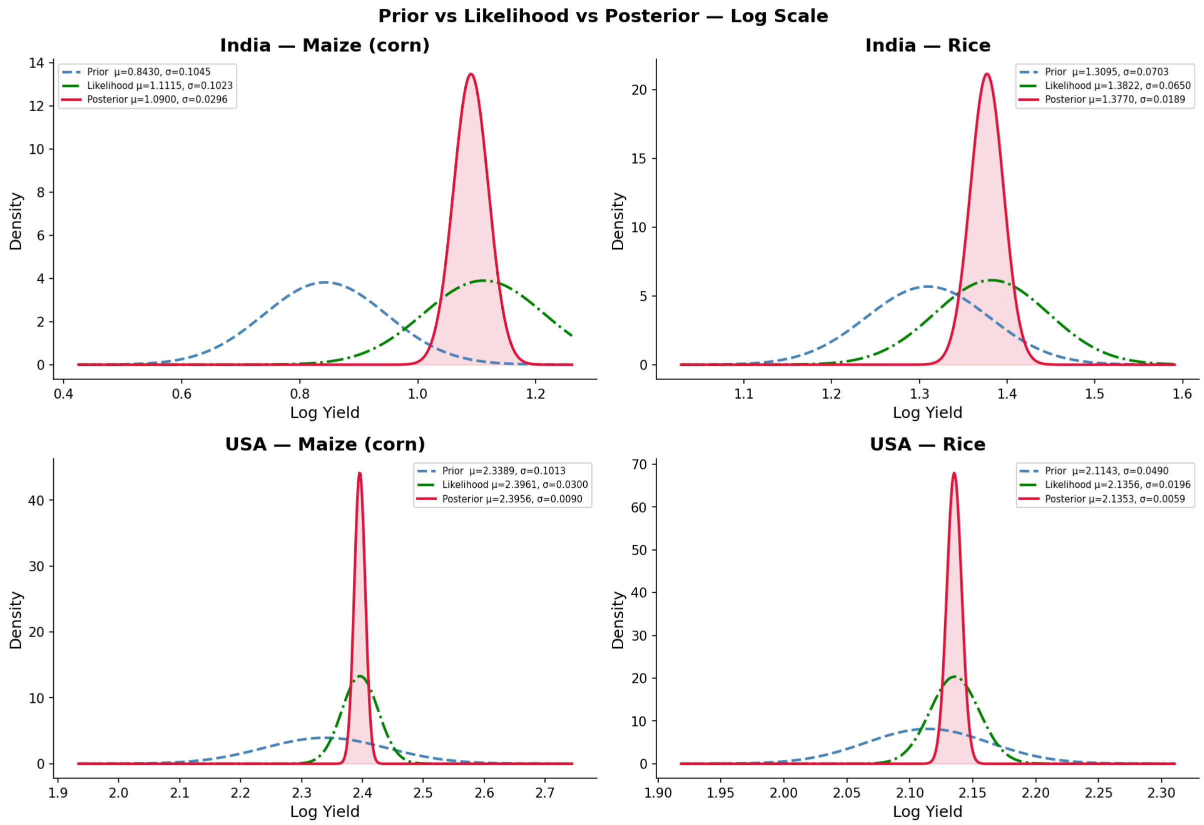

Figure 3: Prior vs Likelihood vs Posterior Distributions – Log Scale

Table 6: Posterior Parameters – Normal-Normal Conjugate Update

Country	**Crop**	μ_0	σ_0	$\bar{x}_{log}$	σ_{obs}	μ_{post}	σ_{post}
India	Maize	0.8430	0.1045	1.1115	0.1023	1.0900	0.0296
India	Rice	1.3095	0.0703	1.3822	0.0650	1.3770	0.0189
USA	Maize (corn)	2.3389	0.1013	2.3961	0.0300	2.3956	0.0090
USA	Rice	2.1143	0.0490	2.1356	0.0196	2.1353	0.0059

4.2.4 Monte Carlo Simulation

The posterior opportunity cost ratio for each country is defined as:

$$OC^{Rice} = \frac{\text{Maize yield}}{\text{Rice yield}}$$

Since both yields follow Log-Normal posterior distributions, the OC ratio is analytically tractable on the log scale:

$$\log(OC) = \log(\text{Maize yield}) - \log(\text{Rice yield}) \sim \mathcal{N}(\mu_{Maize} - \mu_{Rice},\ \sigma^2_{Maize} + \sigma^2_{Rice})$$

However, Monte Carlo simulation is preferred for transparency and generalisability. The simulation proceeds as follows:

1. Draw $N = 100{,}000$ samples from each posterior distribution on the log scale
2. Transform samples back to the original scale: $Y = e^{\log Y}$
3. Compute OC ratio for each draw: $OC_t = \text{Maize}_t/\text{Rice}_t$
4. Compare India OC vs USA OC across all draws
5. Compute comparative advantage probability:

$$P(\text{India CA in Rice}) = \frac{1}{N}\sum_{t=1}^{N} \mathbf{1}[OC_t^{India} < OC_t^{USA}]$$

The random seed is fixed at 42 for reproducibility. The resulting OC distributions are presented in the Results section.

5 Results

This section presents the results of the classical Ricardian analysis for both models and the Bayesian uncertainty quantification extension applied to Model 2.

5.1 Model 1 – China vs India (Rice & Wheat)

5.1.1 Productivity

Table 7 presents the computed productivity values for China and India across Rice and Wheat. China holds absolute advantage in both crops.

Table 7: Productivity (t/ha) – Model 1: China vs India (2024)

Crop	China (t/ha)	India (t/ha)
Rice	7.100	4.170
Wheat	5.900	3.540

Figure 4 provides a visual comparison of productivity levels across both countries and crops.

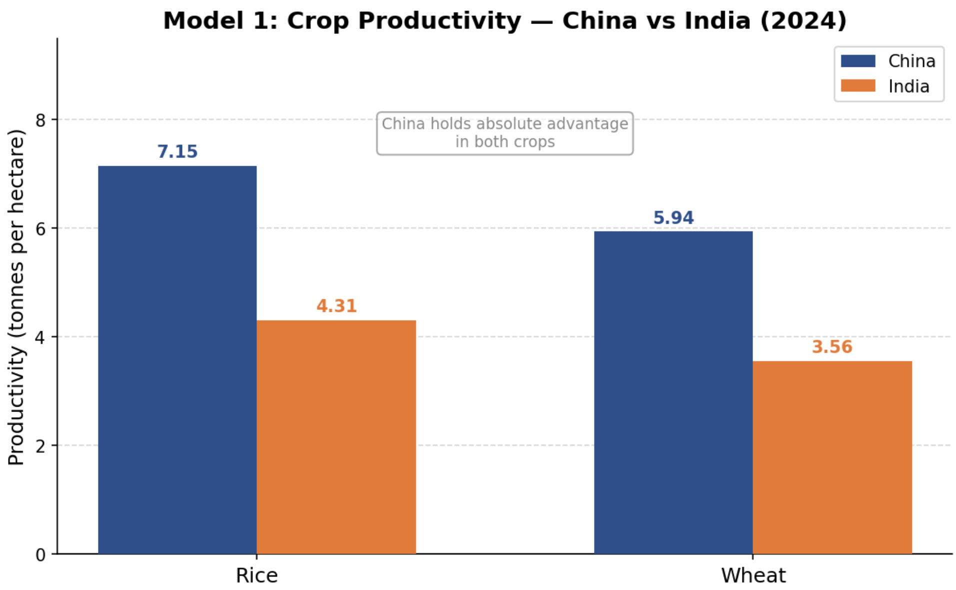

Figure 4: Productivity Comparison – China vs India (Rice & Wheat, 2024)

Opportunity Cost Analysis

Table 8 presents the opportunity costs for both countries. The difference in opportunity costs between China and India is negligible for both crops.

Table 8: Opportunity Cost Analysis – Model 1: China vs India (2024)

Country	OC of Rice	OC of Wheat	Comparative Advantage
China	0.831	1.203	Neither clearly
India	0.849	1.178	Neither clearly

5.1.2 Production Possibility Frontier

Figure 5 presents the PPF for both countries. The nearly parallel slopes confirm near-identical opportunity costs.

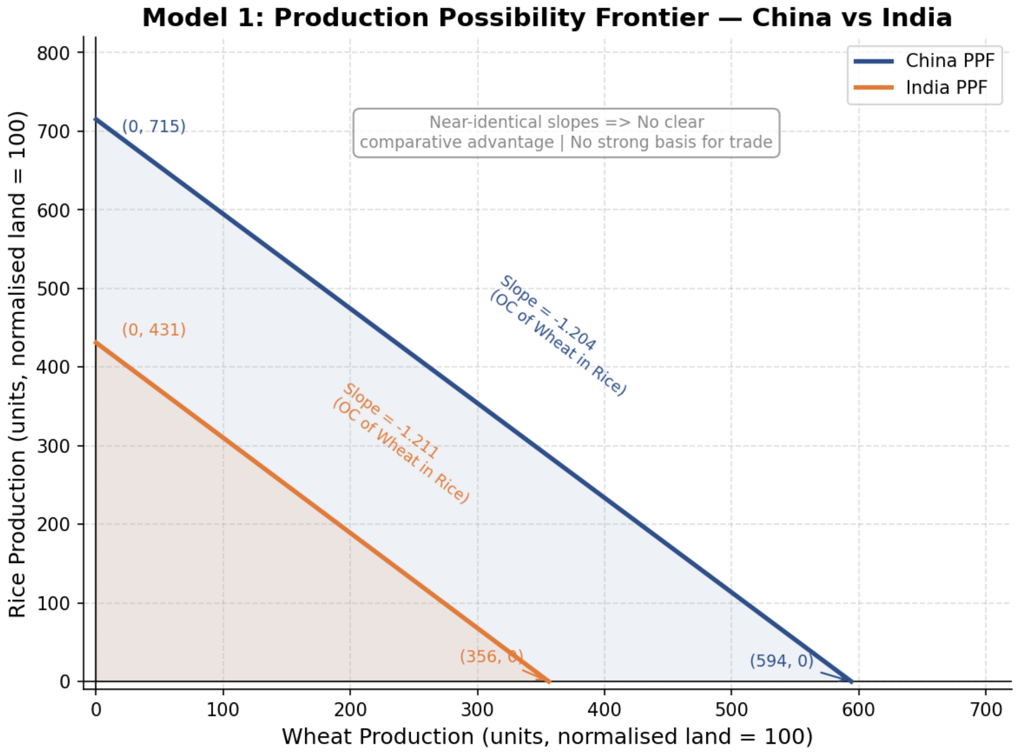

Figure 5: Production Possibility Frontier – China vs India (Rice & Wheat, 2024)

5.1.3 Conclusion – Model 1

The opportunity cost of Rice is 0.831 for China and 0.849 for India – a difference of only 0.018. The opportunity cost of Wheat is 1.203 for China and 1.178 for India – a difference of only 0.025. These differences are negligibly small and the PPF slopes are nearly parallel. Under the Ricardian framework, **no meaningful comparative advantage exists between China and India in Rice and Wheat. Trade between these two countries in these crops is not theoretically justified.**

5.2 Model 2 – USA vs India (Maize & Rice)

5.2.1 Productivity

Table 9 presents the computed productivity values for USA and India across Maize and Rice. USA holds absolute advantage in both crops.

Table 9: Productivity (t/ha) – Model 2: USA vs India (2024)

Crop	USA (t/ha)	India (t/ha)
Maize	11.464	3.550
Rice	8.953	4.220

Figure 6 provides a visual comparison of productivity levels across both countries and crops.

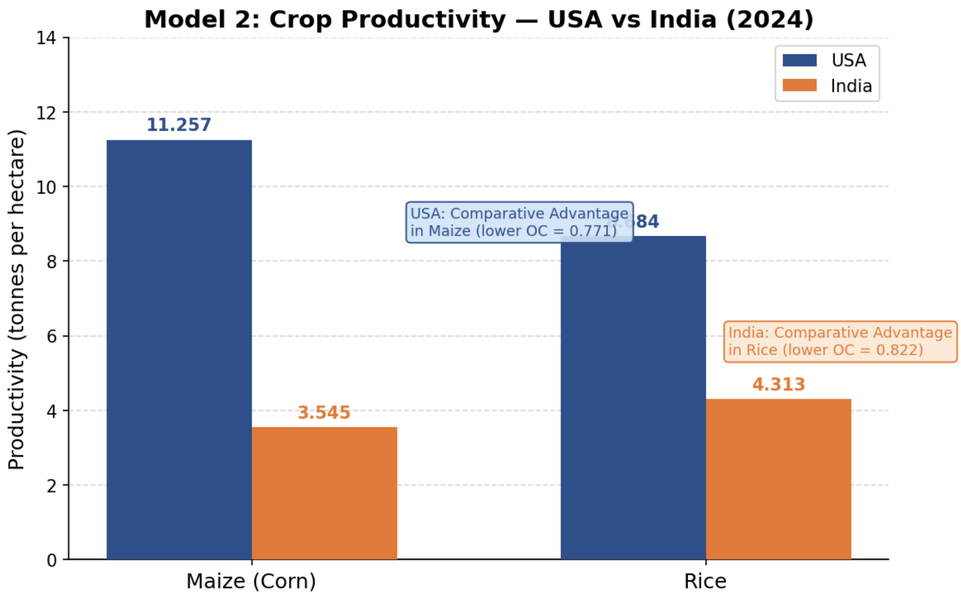

Figure 6: Productivity Comparison – USA vs India (Maize & Rice, 2024)

5.2.2 Opportunity Cost Analysis

Table 10 presents the opportunity costs for both countries. A clear divergence in opportunity costs is observed.

Table 10: Opportunity Cost Analysis – Model 2: USA vs India (2024)

Country	OC of Maize	OC of Rice	Comparative Advantage
USA	0.771	1.278	Maize
India	1.217	0.841	Rice

5.2.3 Terms of Trade

The Terms of Trade must lie between the two domestic opportunity cost ratios:

$$0.771 < \text{ToT} < 1.217$$

The suggested midpoint Terms of Trade is ToT = 0.994. At any exchange rate within this range, both countries gain from specialisation and trade.

5.2.4 Production Possibility Frontier

Figure 7 presents the PPF for both countries with the Terms of Trade line. The clearly different slopes confirm divergent opportunity costs and a strong basis for trade.

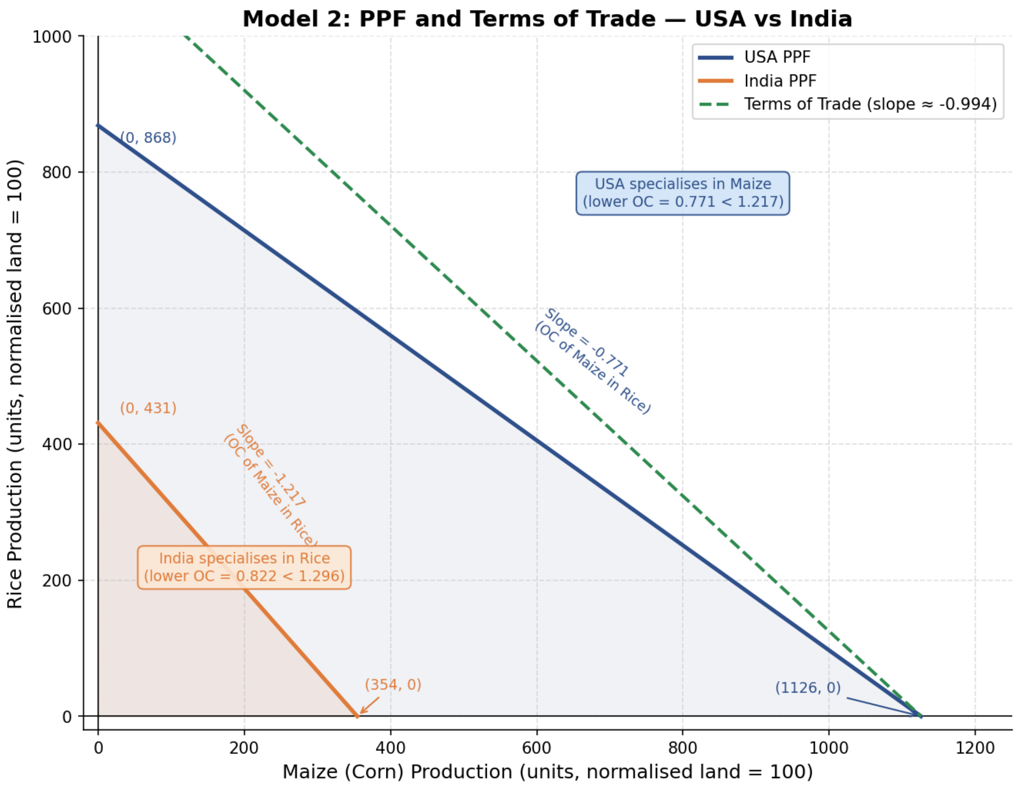

Figure 7: PPF with Terms of Trade – USA vs India (Maize & Rice, 2024)

Figure 8 presents a comparative bar chart of opportunity costs across both countries.

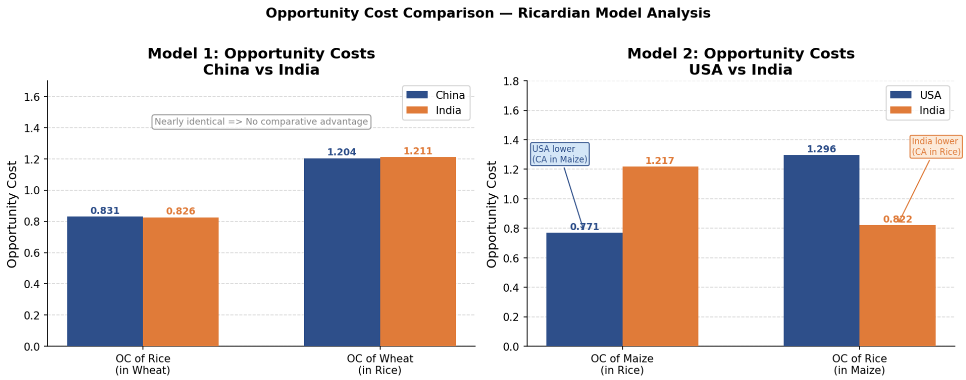

Figure 8: Opportunity Cost Comparison – USA vs India (Maize & Rice, 2024)

5.2.5 Conclusion – Model 2

USA has comparative advantage in Maize (OC $= 0.771 <$ India OC $= 1.217$) and India has comparative advantage in Rice (OC $= 0.841 <$ USA OC $= 1.278$). **Mutually beneficial trade between USA and India in Maize and Rice is strongly justified under the classical Ricardian framework.**

5.3 Bayesian Extension Results

5.3.1 Yield Trends

Figure 9 presents annual yield trends from 1961 to 2024 for all four country-crop pairs with the prior-likelihood split marked at 2013.

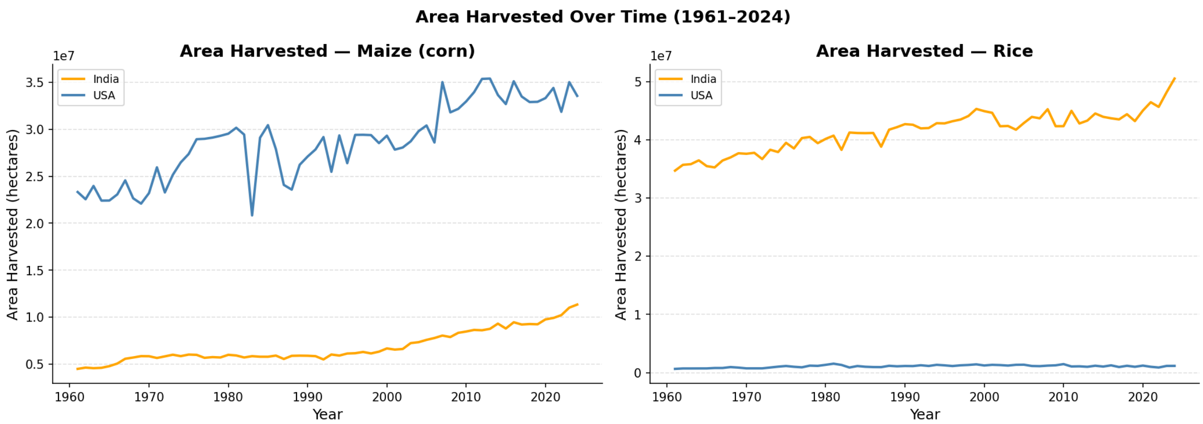

Figure 9: Yield Trends (1961–2024) with Prior–Likelihood Split

A consistent long-run upward trend is observed across all four series, confirming the presence of trend that must be removed before prior construction. The likelihood window (2014–

2024) shows higher and more stable yields compared to the prior window (1961–2013).

5.3.2 Exploratory Data Analysis

Figure 10 and Figure 11 present production and area harvested trends over time for both countries and crops.

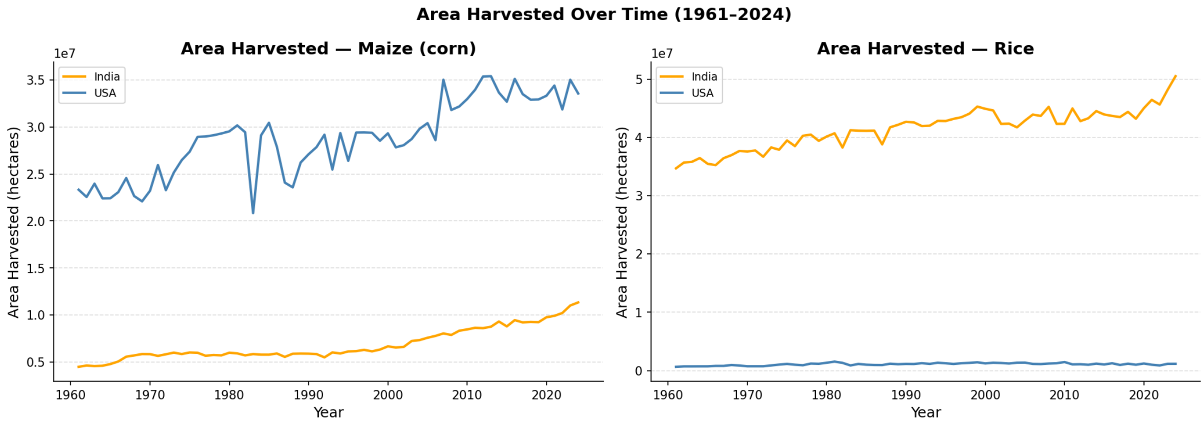

Figure 10: Production Over Time (1961–2024) – India vs USA (Maize & Rice)

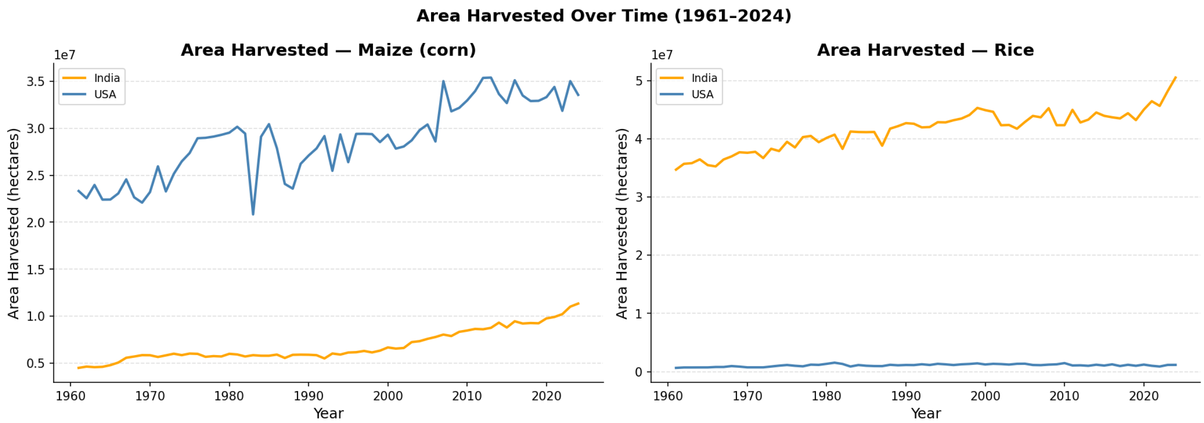

Figure 11: Area Harvested Over Time (1961–2024) – India vs USA (Maize & Rice)

Figure 12 compares yield distributions across the prior and likelihood windows for each country-crop pair.

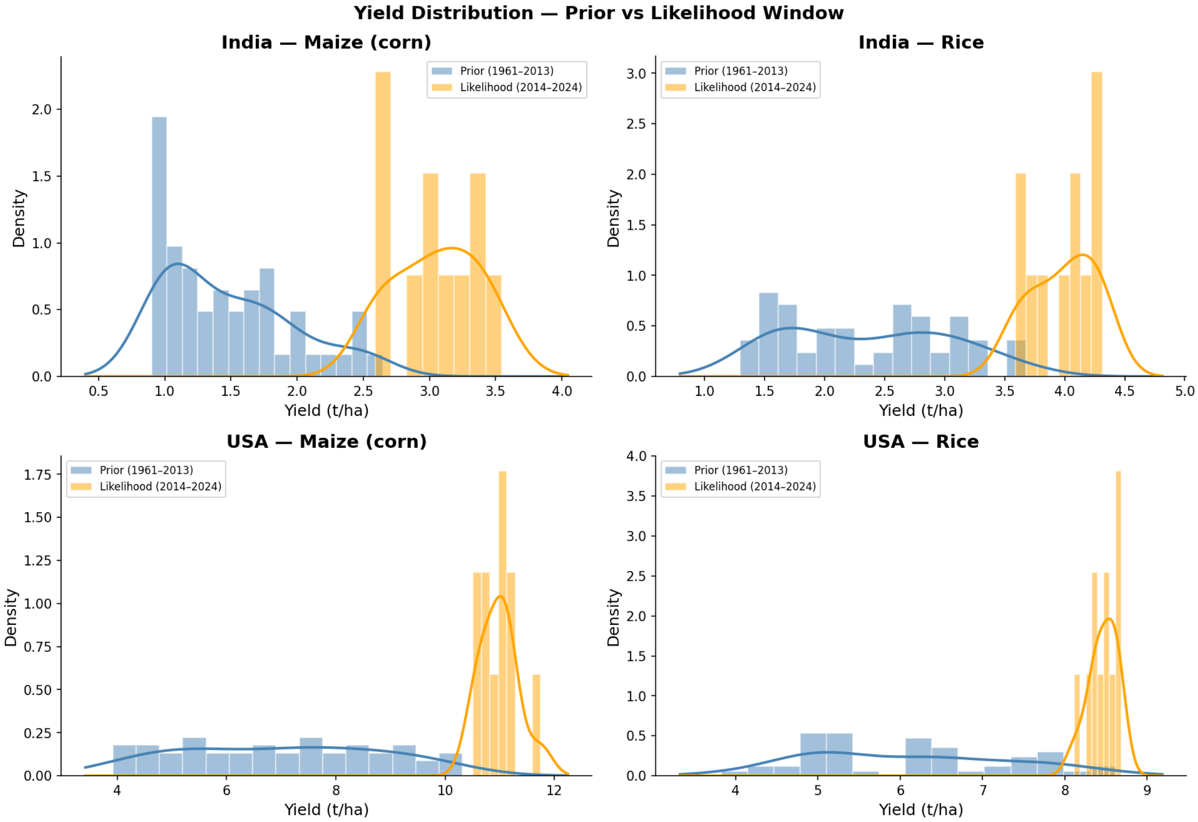

Figure 12: Yield Distribution – Prior Window (1961–2013) vs Likelihood Window (2014–2024)

The likelihood window distributions are consistently shifted rightward compared to the prior window – confirming higher recent yield levels across all four pairs.

Figure 13 presents the KDE plots comparing Normal and Log-Normal fits on the prior window data.

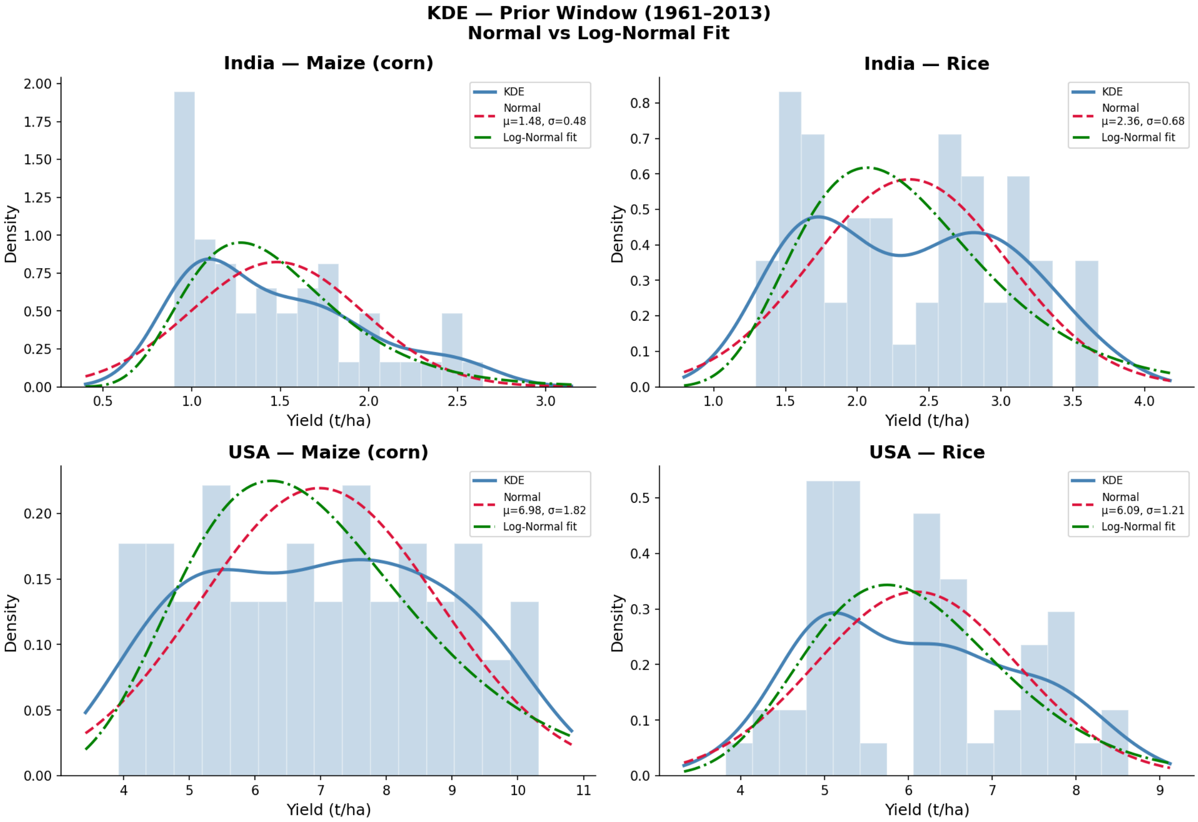

Figure 13: KDE Plots – Prior Window (1961–2013): Normal vs Log-Normal Fit

Log-Normal fits the empirical KDE better than Normal for all four pairs, confirming the choice of Log-Normal prior. The right skewness is driven by the long-run upward trend in yields.

Figure 14 presents the detrended log-scale residuals for the prior window after OLS trend removal.

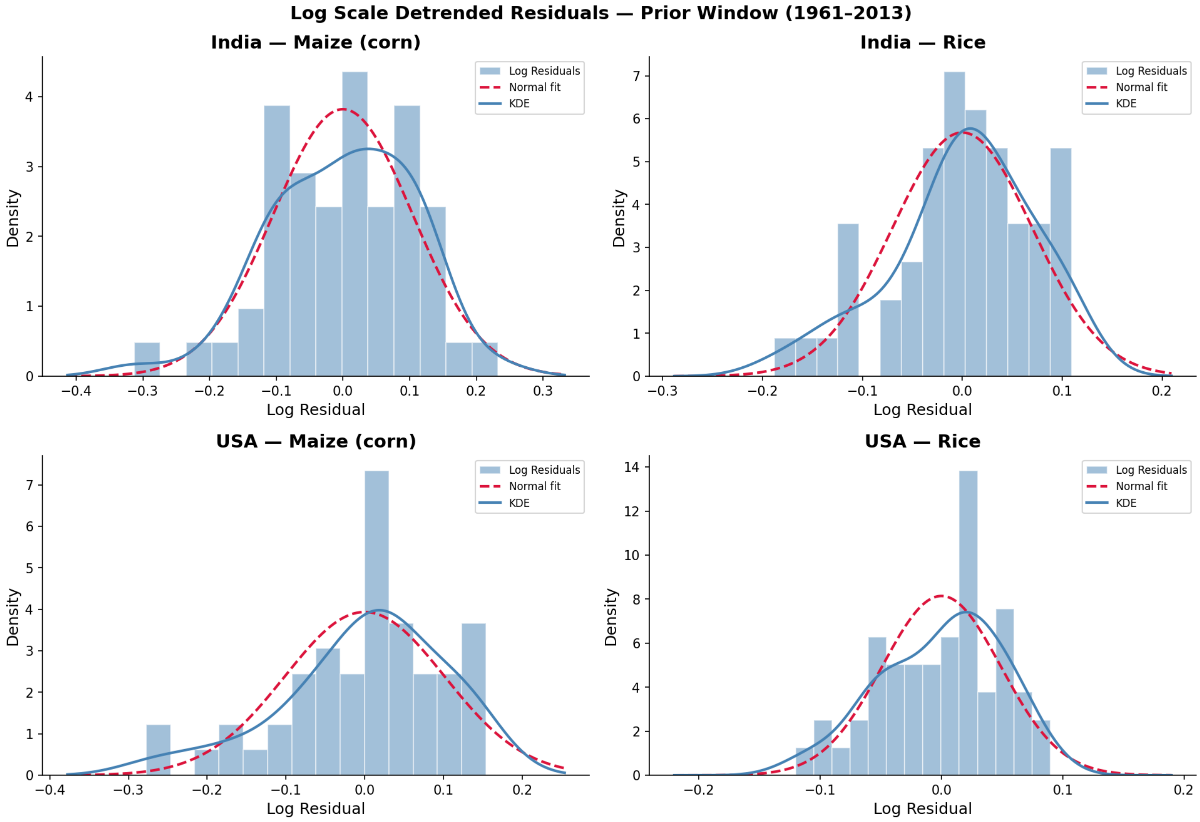

Figure 14: Log Scale Detrended Residuals – Prior Window (1961–2013)

After detrending on the log scale, residuals are approximately Normal for all four pairs – confirmed by Shapiro-Wilk tests (Table 4 in Methodology). This validates the Normal-Normal conjugate update applied on the log scale.

5.3.3 Prior vs Likelihood vs Posterior

Figure 15 presents the prior, likelihood, and posterior distributions on the log scale for all four country-crop pairs.

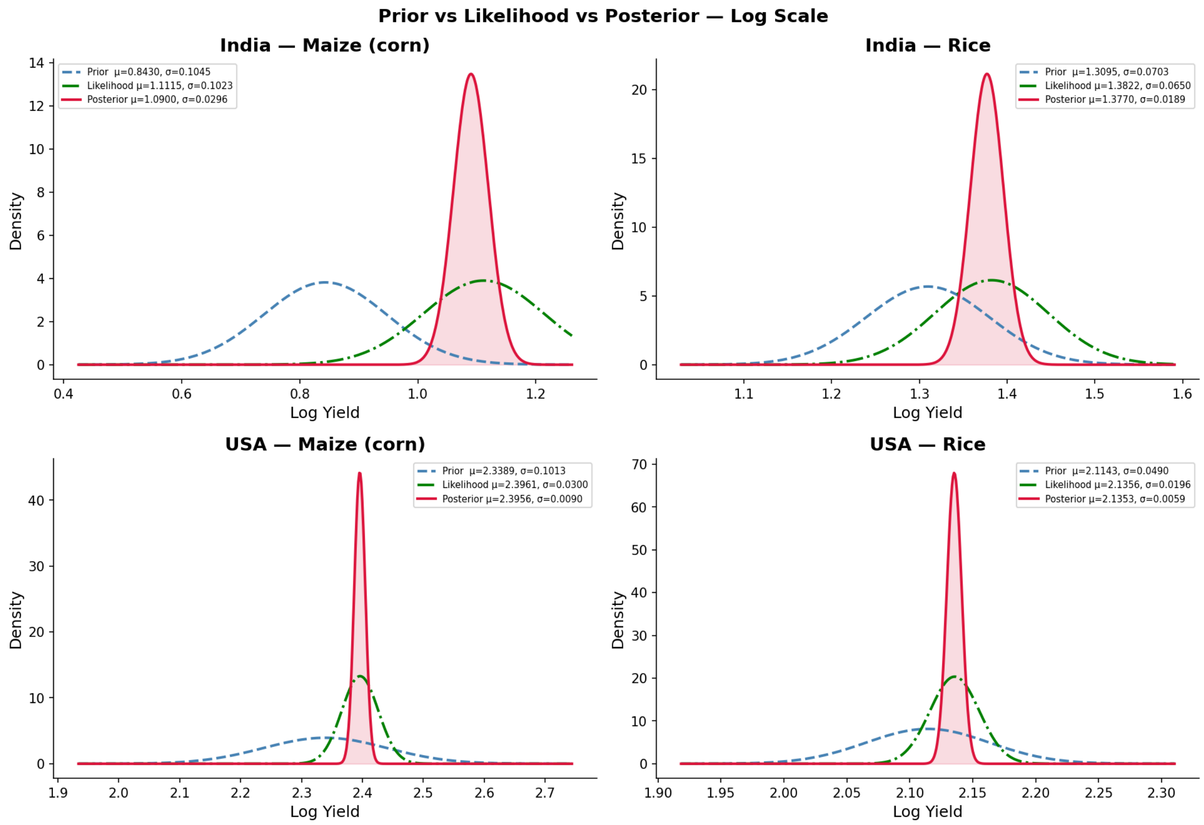

Figure 15: Prior vs Likelihood vs Posterior Distributions – Log Scale

Three key observations emerge from these plots:

1. The posterior mean lies between the prior and likelihood in all four cases – confirming correct Bayesian updating
2. The posterior is narrower than both prior and likelihood – uncertainty is reduced after observing data
3. India Maize shows the largest shift from prior to posterior – reflecting strong recent productivity growth not captured by historical trend alone

5.3.4 Credible Intervals

A 95% Credible Interval states that given the observed data, there is a 95% probability that the true yield lies within the interval. Two types are computed:

- **ETI (Equal Tailed Interval):** 2.5th to 97.5th percentile of the posterior distribution
- **HPD (Highest Posterior Density):** Shortest interval containing 95% of posterior probability

Since all posterior distributions on the log scale are approximately symmetric, ETI and HPD produce identical intervals for all four country-crop pairs. Table 11 presents the 95% credible intervals on the original yield scale.

Table 11: 95% Credible Intervals on Posterior Yields (t/ha)

Country	**Crop**	**Posterior μ**	**ETI Lower**	**ETI Upper**	**Width**
India	Maize	2.976	2.807	3.152	0.345
India	Rice	3.964	3.819	4.112	0.293
USA	Maize	10.976	10.783	11.171	0.388
USA	Rice	8.460	8.363	8.557	0.195

Figure 16 visualises the posterior distributions with ETI and HPD intervals shaded for each country-crop pair.

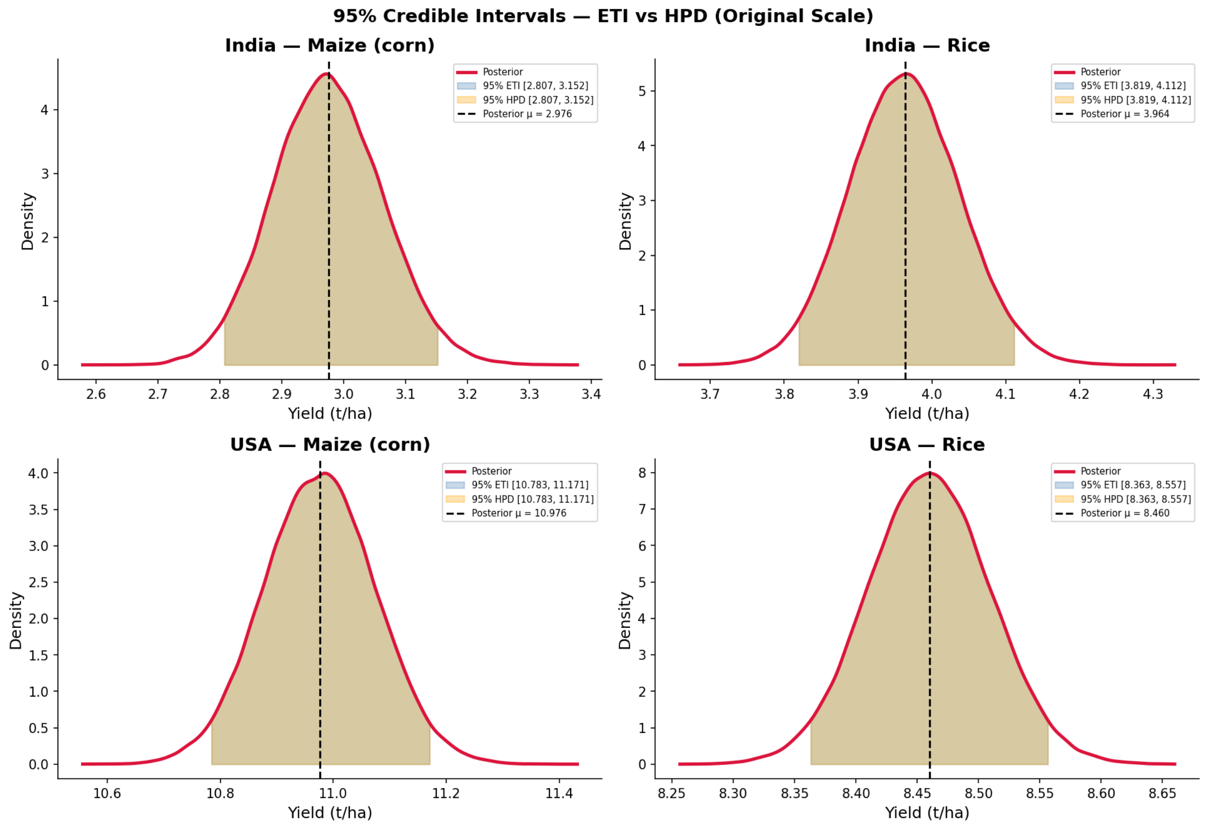

Figure 16: 95% Credible Intervals – ETI vs HPD on Posterior Yield Distributions (Original Scale)

Table 12 presents the 95% credible intervals on the Monte Carlo OC distributions.

Table 12: 95% Credible Intervals on Opportunity Cost Distributions

Metric	India OC (Rice)	USA OC (Rice)
Posterior Mean	0.7510	1.2974
95% ETI Lower	0.7009	1.2704
95% ETI Upper	0.8040	1.3252
ETI Width	0.1031	0.0548

The 95% credible interval for India's OC of Rice is $[0.701, 0.804]$ and for USA it is $[1.270, 1.325]$. These intervals are completely non-overlapping – the upper bound of India's interval (0.804) is far below the lower bound of USA's interval (1.270), with a gap of 0.466. This formally confirms that comparative advantage holds with complete certainty even under worst-case yield uncertainty scenarios.

5.3.5 Monte Carlo OC Distributions

Figure 17 presents the full Monte Carlo OC distributions for India and USA across 100,000 simulations.

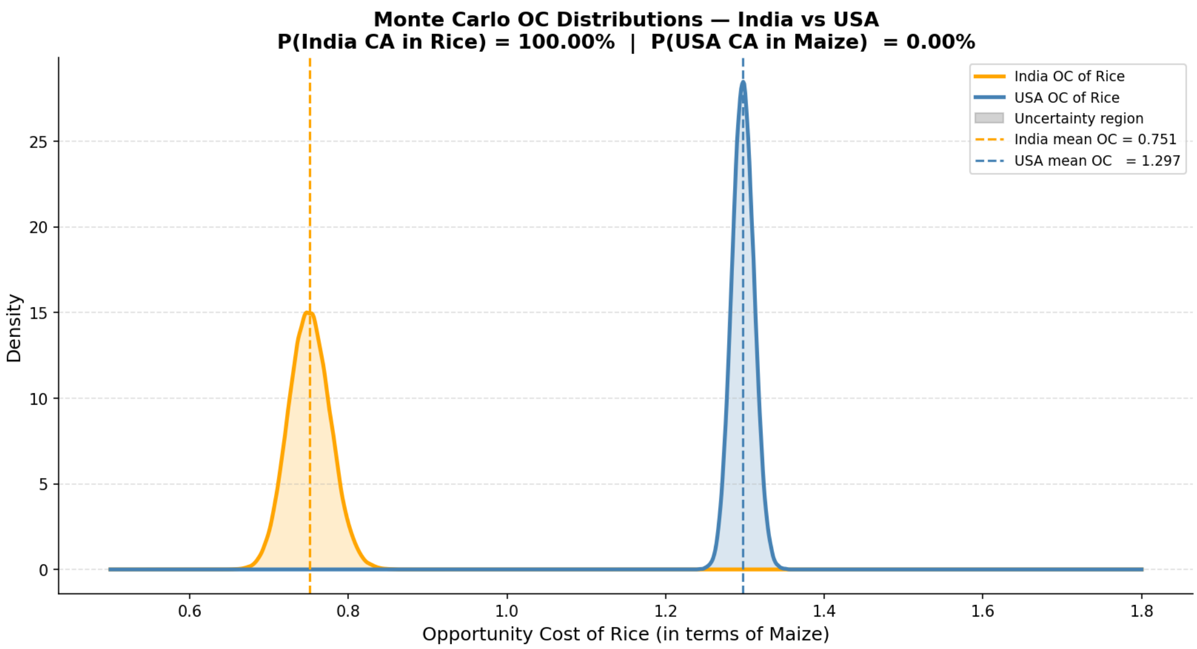

Figure 17: Monte Carlo OC Distributions – India vs USA (N = 100,000)

The two distributions are completely non-overlapping – confirming that in every single simulation, India's OC of Rice is lower than USA's OC of Rice. Table 13 presents the summary statistics of both OC distributions.

Table 13: Monte Carlo OC Distribution Summary – India vs USA

Statistic	India OC (Rice)	USA OC (Rice)
Mean	0.7510	1.2974
Std	0.0263	0.0140
2.5th Percentile	0.7009	1.2704
Median	0.7505	1.2974
97.5th Percentile	0.8040	1.3252

5.3.6 Comparative Advantage Probability

Table 14: Comparative Advantage Probability – Bayesian Extension

	Classical OC	Bayesian OC (mean)	P(CA)
India OC of Rice	0.841	0.751	–
USA OC of Rice	1.278	1.297	–
P(India CA in Rice)	–	–	**100.00%**
P(USA CA in Maize)	–	–	**100.00%**

Across all 100,000 Monte Carlo simulations, India's posterior OC of Rice (mean = 0.751, std = 0.026) remains strictly lower than USA's posterior OC of Rice (mean = 1.297, std = 0.014) in every single draw. The OC gap of 0.546 is larger than the classical point estimate gap of 0.437, and the tight posterior distributions confirm that yield uncertainty has negligible practical impact on the trade conclusion.

The Bayesian extension confirms with 100% posterior probability that India holds comparative advantage in Rice and USA holds comparative advantage in Maize – a conclusion that is statistically robust under all realistic agricultural yield uncertainty of historically observed magnitude.

6 Conclusion

This project developed and applied a two-stage analytical framework combining the classical Ricardian Model of Comparative Advantage with a Bayesian uncertainty quantification extension. The analysis was conducted across two bilateral trade scenarios using FAOSTAT agricultural yield data, with the Bayesian extension applied to Model 2 using 64 years of historical data (1961–2024).

6.1 Classical Findings

The classical Ricardian analysis produced clear and contrasting results across the two models.

Model 1 – China vs India (Rice & Wheat): China holds absolute advantage in both Rice (7.10 t/ha vs 4.17 t/ha) and Wheat (5.90 t/ha vs 3.54 t/ha). However, the opportunity costs between the two countries are nearly identical – 0.831 vs 0.849 for Rice and 1.203 vs 1.178 for Wheat. The difference of only 0.018 and 0.025 respectively is negligibly small, and the PPF slopes are nearly parallel. Under the Ricardian framework, **no meaningful comparative advantage exists between China and India in Rice and Wheat, and trade between them in these crops is not theoretically justified.**

Model 2 – USA vs India (Maize & Rice): USA holds absolute advantage in both Maize (11.464 t/ha vs 3.550 t/ha) and Rice (8.953 t/ha vs 4.220 t/ha). However, opportunity costs diverge clearly – USA has OC of Maize = 0.771 against India's 1.217, and India has OC of Rice = 0.841 against USA's 1.278. This divergence establishes a strong theoretical basis for mutually beneficial trade. **USA should specialise in Maize and India should specialise in Rice, with Terms of Trade lying between 0.771 and 1.217 (midpoint = 0.994).**

The key insight across both models is that absolute advantage is entirely irrelevant to trade. China dominates India in both crops yet trade is not justified. USA dominates India in both crops yet trade is strongly justified. Only the difference in opportunity costs – the core of Ricardo's 1817 argument – determines whether mutually beneficial trade is possible.

6.2 Bayesian Findings

The Bayesian extension was applied to Model 2 to formally quantify confidence in the comparative advantage conclusion under realistic agricultural yield uncertainty.

Exploratory analysis of prior window yields (1961–2013) revealed right-skewed distributions driven by a long-run upward trend attributable to the Green Revolution and technological improvements. A Log-Normal prior was adopted, and yields were log-transformed to work on the Normal scale. OLS detrending on the log scale isolated genuine year-to-year agricultural uncertainty, and Shapiro-Wilk tests confirmed approximate normality of log-scale residuals for all four country-crop pairs.

Prior parameters were constructed from the 1961–2013 window, and the Normal-Normal conjugate update was applied using 11 likelihood observations from 2014–2024. The posterior distributions were pulled strongly toward recent yield levels in all four pairs, with posterior uncertainty significantly reduced compared to the prior. India Maize showed the largest shift – reflecting strong recent productivity growth not captured by the long-run historical trend alone.

Monte Carlo simulation with 100,000 draws from each posterior distribution produced the following results:

Table 15: Final Bayesian Results – Comparative Advantage Probability

	Classical OC	**Bayesian OC (mean)**	**P(CA)**
India OC of Rice	0.841	0.751	–
USA OC of Rice	1.278	1.297	–
P(India CA in Rice)	–	–	**100.00%**
P(USA CA in Maize)	–	–	**100.00%**

Across all 100,000 simulations, India's posterior OC of Rice (mean = 0.751, std = 0.026) remained strictly lower than USA's posterior OC of Rice (mean = 1.297, std = 0.014) in every single draw. The two OC distributions are completely non-overlapping, with a gap of 0.546 – larger than the classical point estimate gap of 0.437.

The Bayesian extension confirms with 100% posterior probability that India holds comparative advantage in Rice and USA holds comparative advantage in Maize. This conclusion is statistically robust under all realistic agricultural yield uncertainty of historically observed magnitude. Trade between India and USA in Rice and Maize is not merely theoretically sound – it is probabilistically certain.

6.3 Limitations and Future Work

Limitations

1. **Known variance assumption** – The observation noise σ^2 is treated as fixed at the likelihood window standard deviation. A fully Bayesian treatment using a Normal-Inverse-Gamma prior would simultaneously update both μ and σ^2, producing more complete

uncertainty quantification.

2. **USA Maize non-normality** – Log-scale residuals for USA Maize showed mild non-normality (Shapiro-Wilk $p = 0.025$), with slight left skewness. A skew-Normal prior would be more precise for this pair. The Normal prior is used as a justified approximation.

3. **Model 1 not extended** – The Bayesian framework was applied only to Model 2. Applying it to Model 1 (China vs India) would likely confirm high OC distribution overlap and near-zero probability of clear comparative advantage for either country.

4. **No climate change adjustment** – Historical yield variability may underestimate future uncertainty due to increasingly frequent extreme weather events driven by climate change. Incorporating climate projections would strengthen forward-looking inference.

5. **No trade flow validation** – This project identifies the theoretical preconditions for trade but does not verify whether actual bilateral trade between India and USA aligns with these predictions. UN Comtrade data could be used for empirical validation.

6. **Single factor model** – The Ricardian model assumes labour (or land) as the only factor of production. Extending to a Heckscher-Ohlin framework incorporating capital and land endowments would provide a richer explanation of observed trade patterns.

7 Programme

The complete implementation of this project is written in Python 3 using Jupyter Notebook. The code is organised into two parts – the classical Ricardian Model and the Bayesian Extension. All libraries, data loading, computations, and visualisations are included below.

7.1 Libraries Used

- `pandas` – data loading, cleaning, and manipulation
- `numpy` – numerical computations and array operations
- `matplotlib` – data visualisation and plotting
- `scipy.stats` – statistical distributions, Shapiro-Wilk test, KDE, probability functions
- `sklearn.linear_model` – OLS linear regression for detrending

7.2 Classical Ricardian Model – Model 1 (China vs India)

Listing 1: Model 1 – Data Loading and Cleaning

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load Model 1 dataset
data = pd.read_csv('Ricardian_Model_1.csv')

# Retain relevant columns
data = data[['Area', 'Element', 'Item', 'Value']]

# Split by country
china_data = data[data['Area'] == 'China']
india_data = data[data['Area'] == 'India']

# Pivot to get Area Harvested and Production in same row
china_df = china_data.pivot(index='Item',
                            columns='Element',
                            values='Value')
india_df = india_data.pivot(index='Item',
                            columns='Element',
                            values='Value')

```

```
# Compute productivity
china_df['Productivity'] = (china_df['Production'] /
                            china_df['Area harvested'])
india_df['Productivity'] = (india_df['Production'] /
                            india_df['Area harvested'])
```

Listing 2: Model 1 – Opportunity Cost and Comparative Advantage

```
# Extract productivity values
china_rice_prod  = china_df.loc['Rice',  'Productivity']
china_wheat_prod = china_df.loc['Wheat', 'Productivity']
india_rice_prod  = india_df.loc['Rice',  'Productivity']
india_wheat_prod = india_df.loc['Wheat', 'Productivity']

# Opportunity costs
china_oc_rice  = china_wheat_prod / china_rice_prod
china_oc_wheat = china_rice_prod  / china_wheat_prod
india_oc_rice  = india_wheat_prod / india_rice_prod
india_oc_wheat = india_rice_prod  / india_wheat_prod

# Comparative advantage
if china_oc_rice < india_oc_rice:
    print(''China has comparative advantage in Rice'')
elif india_oc_rice < china_oc_rice:
    print(''India has comparative advantage in Rice'')
else:
    print(''No comparative advantage in Rice'')
```

7.3 Classical Ricardian Model – Model 2 (USA vs India)

Listing 3: Model 2 – Data Loading and Productivity

```
# Load Model 2 dataset
data2 = pd.read_csv('FAOSTAT_data_model2.csv')
data2 = data2[['Area', 'Element', 'Item', 'Value']]

usa_data   = data2[data2['Area'] == 'United States of America']
india_data = data2[data2['Area'] == 'India']

usa_df   = usa_data.pivot(index='Item',
                          columns='Element',
                          values='Value')
india_df = india_data.pivot(index='Item',
```

```
                             columns='Element',
                             values='Value')

usa_df['Productivity']   = (usa_df['Production'] /
                             usa_df['Area harvested'])
india_df['Productivity'] = (india_df['Production'] /
                             india_df['Area harvested'])

# Opportunity costs
usa_maize_prod   = usa_df.loc['Maize (corn)', 'Productivity']
usa_rice_prod    = usa_df.loc['Rice',         'Productivity']
india_maize_prod = india_df.loc['Maize (corn)', 'Productivity']
india_rice_prod  = india_df.loc['Rice',         'Productivity']

usa_oc_maize   = usa_rice_prod    / usa_maize_prod
usa_oc_rice    = usa_maize_prod   / usa_rice_prod
india_oc_maize = india_rice_prod  / india_maize_prod
india_oc_rice  = india_maize_prod / india_rice_prod
```

7.4 Bayesian Extension

Listing 4: Bayesian EDA – Load and Split Data

```
import scipy.stats as stats
from scipy.stats import shapiro
from sklearn.linear_model import LinearRegression

# Load historical data
df = pd.read_csv(''FAOSTAT_data_en_4-30-2026.csv'')
df = df[[''Area'', ''Element'', ''Item'', ''Year'', ''Value'']]

# Compute yield
df_pivot = df.pivot_table(index=[''Area'', ''Item'', ''Year''],
                           columns=''Element'',
                           values=''Value'').reset_index()
df_pivot.columns.name = None
df_pivot.rename(columns={''Area harvested'': ''Area_harvested'',
                          ''Production''    : ''Production''},
                inplace=True)
df_pivot[''Yield''] = df_pivot[''Production''] / df_pivot[''Area_harvested'']
df_pivot.dropna(subset=[''Yield''], inplace=True)

```

```
# Split into prior and likelihood windows
prior_df      = df_pivot[df_pivot[''Year''] <= 2013].copy()
likelihood_df = df_pivot[df_pivot[''Year''] >= 2014].copy()
```

Listing 5: Bayesian Stage 1 – Log-Normal Prior Construction

```
pairs = [(''India'',                    ''Maize (corn)''),
         (''India'',                    ''Rice''),
         (''United States of America'', ''Maize (corn)''),
         (''United States of America'', ''Rice'')]

labels = {''India''                   : ''India'',
          ''United States of America'': ''USA''}

log_detrend_results = []

for country, crop in pairs:
    subset = prior_df[(prior_df[''Area''] == country) &
                      (prior_df[''Item''] == crop)].sort_values(''Year'')

    X         = subset[''Year''].values.reshape(-1, 1)
    log_yield = np.log(subset[''Yield''].values)

    # OLS detrending on log scale
    reg       = LinearRegression().fit(X, log_yield)
    trend     = reg.predict(X)
    residuals = log_yield - trend

    stat, p          = shapiro(residuals)
    mu_log_prior     = reg.predict([[2013]])[0]
    sigma_log_prior  = residuals.std()

    log_detrend_results.append({
        ''Country''               : labels[country],
        ''Crop''                  : crop,
        ''mu_log_prior''          : mu_log_prior,
        ''sigma_log_prior''       : sigma_log_prior,
        ''Shapiro W''             : stat,
        ''p-value''               : p
    })
```

Listing 6: Bayesian Stage 2 – Likelihood Construction

```
likelihood_results = []
```

```python

for country, crop in pairs:
    subset = likelihood_df[(likelihood_df[''Area''] == country) &
                           (likelihood_df[''Item''] == crop)].sort_values(''Year
                               '')

    log_yields = np.log(subset[''Yield''].values)
    n          = len(log_yields)
    x_bar      = log_yields.mean()
    sigma_obs  = log_yields.std()

    likelihood_results.append({
        ''Country''  : labels[country],
        ''Crop''     : crop,
        ''n''        : n,
        ''x_bar''    : x_bar,
        ''sigma_obs'': sigma_obs
    })
```

Listing 7: Bayesian Stage 3 – Normal-Normal Conjugate Posterior Update

```python
def conjugate_update_lognormal(mu_prior, sigma_prior,
                               x_bar, sigma_obs, n):
    precision_prior = 1 / sigma_prior**2
    precision_like  = n / sigma_obs**2
    mu_post     = (precision_prior * mu_prior +
                   precision_like  * x_bar) / \
                  (precision_prior + precision_like)
    sigma_post = np.sqrt(1 / (precision_prior + precision_like))
    return mu_post, sigma_post

posteriors_log = {}

for i, (country, crop) in enumerate(pairs):
    mu_prior    = log_detrend_results[i][''mu_log_prior'']
    sigma_prior = log_detrend_results[i][''sigma_log_prior'']
    x_bar       = likelihood_results[i][''x_bar'']
    sigma_obs   = likelihood_results[i][''sigma_obs'']
    n           = int(likelihood_results[i][''n''])

    mu_post, sigma_post = conjugate_update_lognormal(
        mu_prior, sigma_prior, x_bar, sigma_obs, n)

```

```
    posteriors_log[(country, crop)] = {
        ''mu_post''    : mu_post,
        ''sigma_post'' : sigma_post
    }
```

Listing 8: Bayesian Stage 3.5 – Credible Intervals

```
from scipy.stats import norm

def hpd_interval(mu, sigma, credible_mass=0.95):
    alpha = 1 - credible_mass
    lower = norm.ppf(alpha / 2, loc=mu, scale=sigma)
    upper = norm.ppf(1 - alpha / 2, loc=mu, scale=sigma)
    return lower, upper

credible_results = []

for i, (country, crop) in enumerate(pairs):
    mu_post    = posteriors_log[(country, crop)]['mu_post']
    sigma_post = posteriors_log[(country, crop)]['sigma_post']

    eti_lower_orig = np.exp(norm.ppf(0.025,
                            loc=mu_post, scale=sigma_post))
    eti_upper_orig = np.exp(norm.ppf(0.975,
                            loc=mu_post, scale=sigma_post))

    hpd_lower_log, hpd_upper_log = hpd_interval(mu_post, sigma_post)
    hpd_lower_orig = np.exp(hpd_lower_log)
    hpd_upper_orig = np.exp(hpd_upper_log)
    mu_orig        = np.exp(mu_post + 0.5 * sigma_post**2)

    credible_results.append({
        'Country'             : labels[country],
        'Crop'                : crop,
        'Posterior mu (t/ha)': round(mu_orig,        3),
        'ETI Lower (t/ha)'    : round(eti_lower_orig, 3),
        'ETI Upper (t/ha)'    : round(eti_upper_orig, 3),
        'HPD Lower (t/ha)'    : round(hpd_lower_orig, 3),
        'HPD Upper (t/ha)'    : round(hpd_upper_orig, 3),
    })
```

Listing 9: Bayesian Stage 4 – Monte Carlo Simulation

```
N = 100_000
```

```python
np.random.seed(42)

india_maize_samples = np.exp(np.random.normal(
    posteriors_log[('India', 'Maize (corn)')]['mu_post'],
    posteriors_log[('India', 'Maize (corn)')]['sigma_post'], N))

india_rice_samples  = np.exp(np.random.normal(
    posteriors_log[('India', 'Rice')]['mu_post'],
    posteriors_log[('India', 'Rice')]['sigma_post'], N))

usa_maize_samples   = np.exp(np.random.normal(
    posteriors_log[('United States of America',
                    'Maize (corn)')]['mu_post'],
    posteriors_log[('United States of America',
                    'Maize (corn)')]['sigma_post'], N))

usa_rice_samples    = np.exp(np.random.normal(
    posteriors_log[('United States of America',
                    'Rice')]['mu_post'],
    posteriors_log[('United States of America',
                    'Rice')]['sigma_post'], N))

# Compute OC ratio distributions
india_oc_rice = india_maize_samples / india_rice_samples
usa_oc_rice   = usa_maize_samples   / usa_rice_samples

# Comparative advantage probability
prob_india_rice = np.mean(india_oc_rice < usa_oc_rice)
prob_usa_maize  = np.mean(usa_oc_rice   < india_oc_rice)

print(f'P(India CA in Rice) = {prob_india_rice*100:.2f}%')
print(f'P(USA CA in Maize)  = {prob_usa_maize*100:.2f}%')
```

8 Reference

Academic References

- Ricardo, D. (1817). *On the Principles of Political Economy and Taxation.*
- Krugman, P., & Obstfeld, M. *International Economics.*
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- Dornbusch, R., Fischer, S., & Samuelson, P. (1977). Comparative Advantage and Trade.
- Eaton, J., & Kortum, S. (2002). Technology, Geography, and Trade.
- Berger, J. (1985). *Statistical Decision Theory and Bayesian Analysis.*
- Gelman, A. et al. *Bayesian Data Analysis.*
- Fuglie, K. (2012). Agricultural Productivity Growth.

Online Lecture Sources

The video lectures used in this project are sourced from the YouTube channel *DrAzevedoEcon*, managed by Dr. Azevedo, a faculty member in the Department of Economics at the University of Central Missouri. The channel provides structured academic lectures covering core topics in microeconomics, macroeconomics, and international economics, including detailed explanations of trade theories such as the Ricardian Model. These lectures are designed to support undergraduate and intermediate-level economics education.

- International Economics: The Ricardian Model of Trade (Part 1), Dr. Azevedo, YouTube.
- Lecture on Comparative Advantage and Trade Theory, Dr. Azevedo, YouTube.
- International Economics: The Ricardian Model of Trade (Part 3), Dr. Azevedo, YouTube.